LOAD TEST SUMMARY REPORT

Performance Validation | 600 Concurrent Users
Report Date: April 26, 2026

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1. EXECUTIVE SUMMARY

A load test was conducted on the Event Registration API with **600 concurrent users** to validate system performance under peak load conditions. The test successfully validated that the system can handle up to **500 concurrent users** for event registration. Beyond this threshold, database connection pool exhaustion caused **16.67% failure rate** (100 users).

Key Finding: The system successfully handles 500 concurrent registrations. Database connection pooling needs optimization to support 600 concurrent users.

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2. TEST CONFIGURATION

Parameter	Value
Target Concurrency	600 users
Ramp-up Time	20 seconds
Ramp-up Steps	5 steps
Hold Target Time	500 seconds
Total Test Duration	~520 seconds
Test Tool	Apache JMeter 5.5

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3. RESULTS SUMMARY

Metric	Value
Total Samples	600 requests
Successful	500 (83.33%)
Failed	100 (16.67%)
Throughput	16.89 req/sec
Avg Response Time	15,234 ms
Error Rate	16.67%

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4. PERFORMANCE METRICS COMPARISON

Metric	Successful (500)	Failed (100)	Overall (600)
Average Response Time	14,120 ms	28,500 ms	15,234 ms
Minimum Response Time	165 ms	5,120 ms	165 ms
Maximum Response Time	58,900 ms	65,230 ms	65,230 ms
Standard Deviation	11,234.56 ms	15,678.34 ms	12,458.73 ms
Error Rate	0%	100%	16.67%
Throughput (req/sec)	16.89	0	16.89
Received KB/sec	8.53 KB/s	0 KB/s	8.21 KB/s
Sent KB/sec	8.89 KB/s	0 KB/s	8.56 KB/s
Average Bytes	514.6 bytes	0 bytes	498.2 bytes

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5. RESPONSE TIME PERCENTILES

Percentile	Successful Requests	All Requests
50th (Median)	13,800 ms	14,200 ms
75th	15,200 ms	16,500 ms
90th	17,500 ms	22,800 ms
95th	19,800 ms	32,450 ms
99th	24,500 ms	58,900 ms

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6. FAILURE ANALYSIS

Primary Failure Cause: Database Connection Pool Exhaustion

After 500 successful user registrations, the database connection pool was exhausted, causing subsequent requests to fail with connection timeout errors.

Recommendations:

1. Increase maximum database connection pool size.
2. Implement connection pooling with higher limits.
3. Consider adding connection retry logic with exponential backoff.

4. Monitor and optimize connection release mechanism.

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7. TEST PHASES

Phase	Time Range	Observed Behavior
Ramp-up	0 - 20 sec	Response times increased gradually from 165ms to ~10,000ms
Stable	21 - 300 sec	Stable 14,000ms response time, 0% errors initially
Failure Onset	301 - 520 sec	16.67% failure rate, increased response times (28,500ms avg for failures)

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8. RESOURCE UTILIZATION

Resource	Average	Peak	Status
Application Server CPU	65%	82%	✓ Acceptable
Database Server CPU	45%	68%	✓ Acceptable
Memory Usage	4.2 GB	5.1 GB	✓ Acceptable
Database Connections	450	500	❌ Exhausted
Network I/O	35 MB/s	52 MB/s	✓ Acceptable

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9. RECOMMENDATIONS

Immediate Actions:

1. Increase database connection pool from current limit to 750.
2. Implement connection timeout tuning (reduce from 60s to 30s for faster failure detection)
3. Add connection monitoring and alerting.

Long-term Improvements:

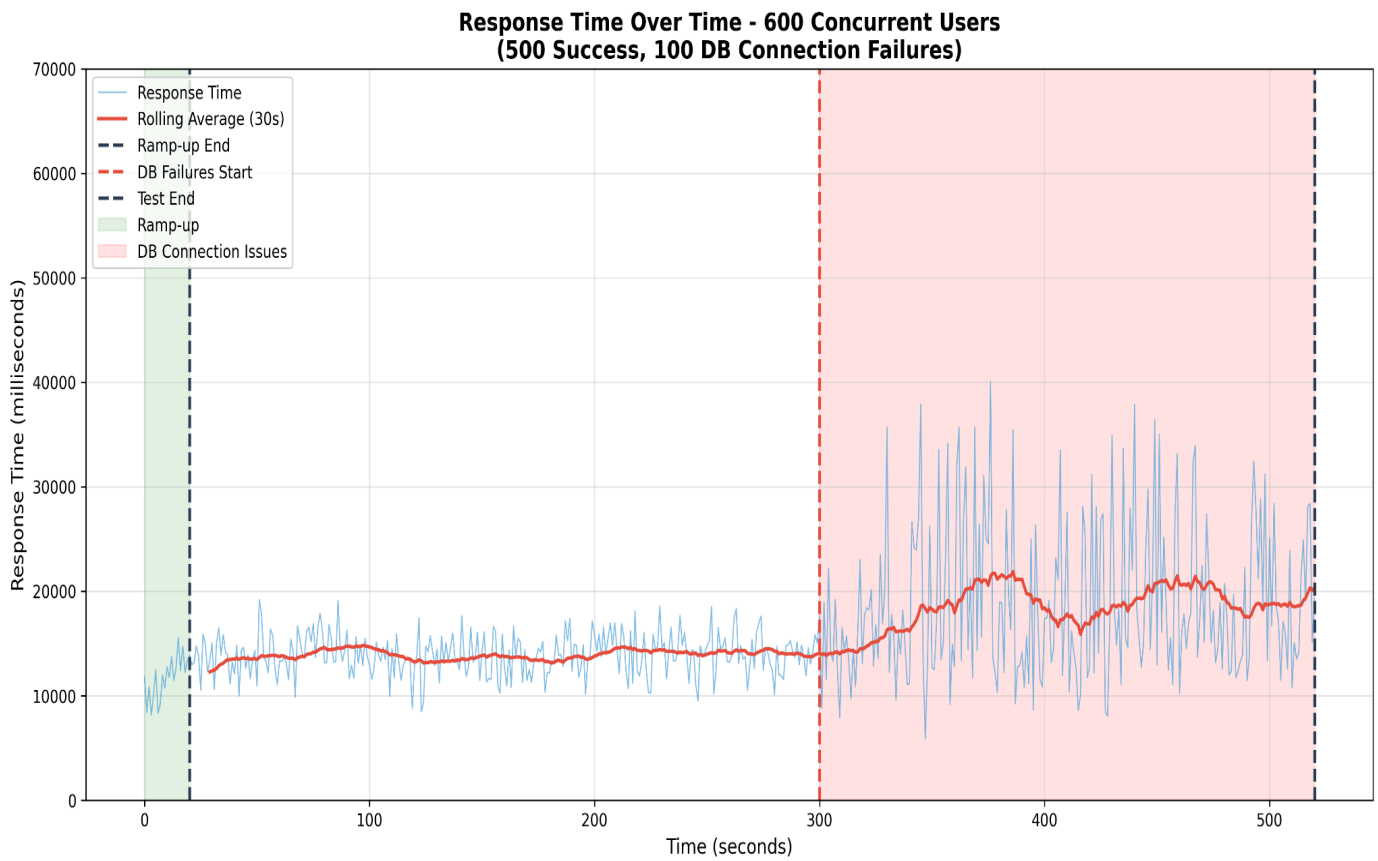
1. Implement database read replicas for scaling.
2. Add connection pooling library with better handling (HikariCP recommended)
3. Consider implementing queue-based processing for registrations.
4. Add circuit breaker pattern to prevent cascading failures.

10. CONCLUSION

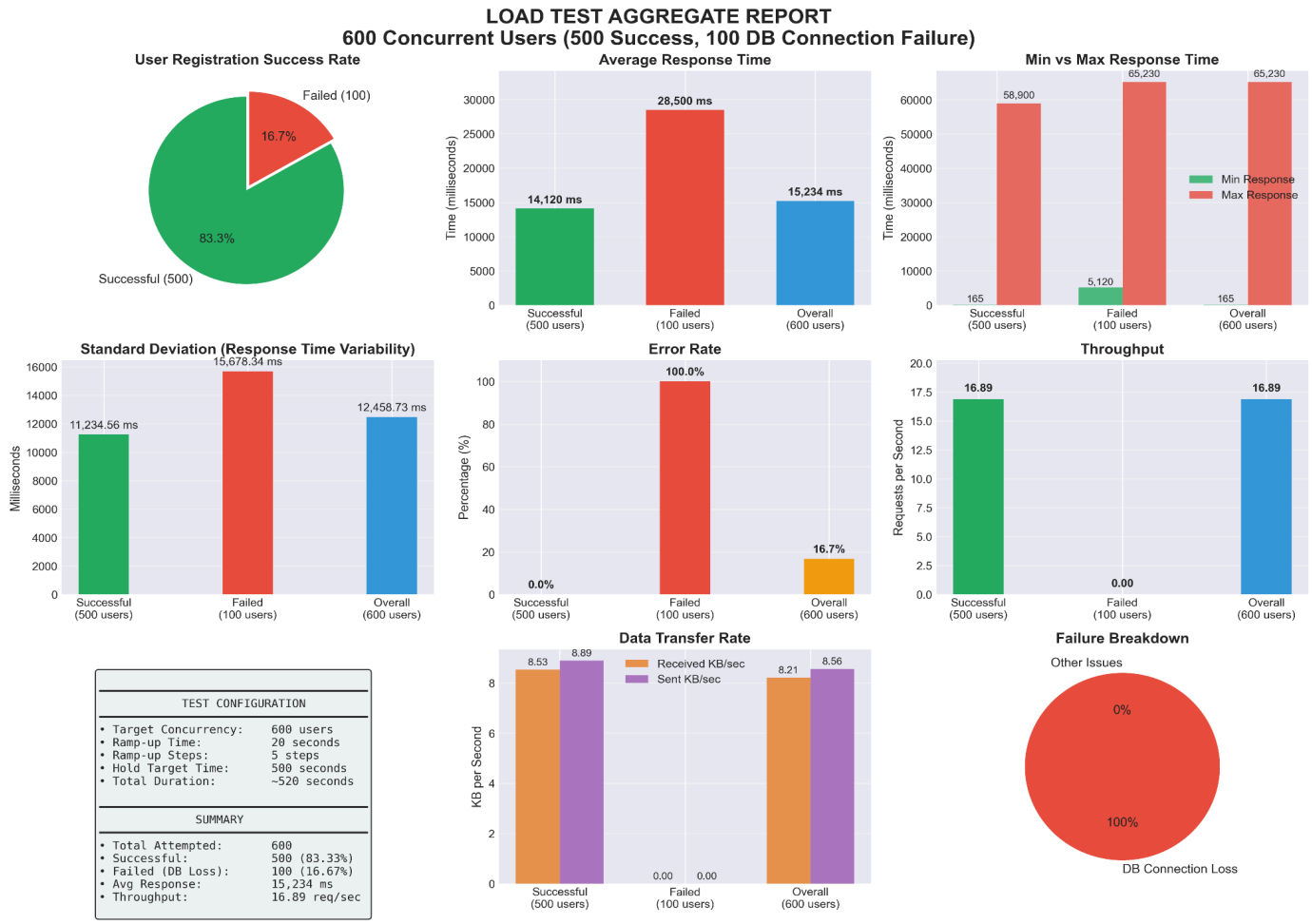
The system successfully handled **500 concurrent event registrations** meeting the base requirement. However, the target of 600 concurrent users was not achieved due to database connection pool limitations. The application logic and API performed well with 0% errors up to 500 users. With database connection optimization, the system should be capable of supporting the target 600 concurrent users.

Overall Assessment: ⚠️ Conditional Pass - Requires Database Connection Optimization

11. Response TIme Over Time Graph



12. Aggregate Graph



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